



ANDROID STATIC ANALYSIS REPORT



 Uncrackable Level 2 (1.0)

File Name: UnCrackable-Level2_1_.apk

Package Name: owasp.mstg.uncrackable2

Scan Date: April 25, 2026, 1:46 p.m.

App Security Score: **65/100 (LOW RISK)**

Grade:



FINDINGS SEVERITY

 HIGH	 MEDIUM	 INFO	 SECURE	 HOTSPOT
1	3	0	2	0

FILE INFORMATION

File Name: UnCrackable-Level2_1_.apk

Size: 0.86MB

MD5: f9f31bfe8df27eb74cdb8f14fd7c85df

SHA1: f9b4d88a7e0c99f3007579622be9255ca26833ff

SHA256: 4c7980ef1f6cc29508f38417c2a5b5b7721eeb4a3d7f00a0c1a08f5192fa6ddd

APP INFORMATION

App Name: Uncrackable Level 2

Package Name: owasp.mstg.uncrackable2

Main Activity: sg.vantagepoint.uncrackable2.MainActivity

Target SDK: 28

Min SDK: 19

Max SDK:

Android Version Name: 1.0

Android Version Code: 1

APP COMPONENTS

Activities: 1

Services: 0

Receivers: 0

Providers: 0

Exported Activities: 0

Exported Services: 0

Exported Receivers: 0

Exported Providers: 0

CERTIFICATE INFORMATION

Binary is signed

v1 signature: True

v2 signature: True

v3 signature: False

v4 signature: False

X.509 Subject: C=NL, L=Amsterdam, O=OWASP, OU=OMTG, CN=Jeroen Willemsen

Signature Algorithm: rsassa_pkcs1v15

Valid From: 2018-08-27 21:01:01+00:00

Valid To: 2043-08-21 21:01:01+00:00

Issuer: C=NL, L=Amsterdam, O=OWASP, OU=OMTG, CN=Jeroen Willemsen

Serial Number: 0x71861047

Hash Algorithm: sha256

md5: 1061e3c49001fcc5f079325278df94ca

sha1: 13e49a39e628817cbbde2881f4646c2a0b22adb2

sha256: cbe68ab87d64341c3fc78119dc1229e8aa86f93fdb2c33aa61252fd491485b1d

sha512: 30ce347be73baca32bd740262d54007ead7ac524f4b62dfc13019ca2b4a0b08820f2ef8da9e9c77d6bdc14567edfbed1500069dd178b7e0432ada053963db1cd

PublicKey Algorithm: rsa

Bit Size: 2048

Fingerprint: e3111e7eef44e1fff8763dfb4a27833d9ec4fc9debf2cab052ac0194ca25bfd0

Found 1 unique certificates

APKID ANALYSIS

FILE	DETAILS	
classes.dex	FINDINGS	DETAILS
	Anti Debug Code	Debug.isDebuggerConnected() check
	Anti-VM Code	Build.TAGS check
	Compiler	dx r8 without marker (suspicious)

NETWORK SECURITY

NO	SCOPE	SEVERITY	DESCRIPTION
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CERTIFICATE ANALYSIS

HIGH: 0 | WARNING: 1 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate

TITLE	SEVERITY	DESCRIPTION
Application vulnerable to Janus Vulnerability	warning	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.

Q MANIFEST ANALYSIS

HIGH: 1 | WARNING: 1 | INFO: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 4.4-4.4.4, [minSdk=19]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.

</> CODE ANALYSIS

HIGH: 0 | WARNING: 1 | INFO: 0 | SECURE: 1 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	This App may request root (Super User) privileges.	warning	CWE: CWE-250: Execution with Unnecessary Privileges OWASP MASVS: MSTG-RESILIENCE-1	sg/vantagepoint/a/b.java
2	This App may have root detection capabilities.	secure	OWASP MASVS: MSTG-RESILIENCE-1	sg/vantagepoint/a/b.java

SHARED LIBRARY BINARY ANALYSIS

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
1	armeabi-v7a/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
2	x86/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
3	arm64-v8a/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
4	x86_64/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
5	armeabi-v7a/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
6	x86/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
7	arm64-v8a/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
8	x86_64/libfoo.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NIAP ANALYSIS v1.3

NO	IDENTIFIER	REQUIREMENT	FEATURE	DESCRIPTION
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⚙️ ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	0/25	
Other Common Permissions	0/44	

Malware Permissions:

Top permissions that are widely abused by known malware.

Other Common Permissions:

Permissions that are commonly abused by known malware.

! OFAC SANCTIONED COUNTRIES

This app may communicate with the following OFAC sanctioned list of countries.

DOMAIN	COUNTRY/REGION
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🔍 DOMAIN MALWARE CHECK

DOMAIN	STATUS	GEOLOCATION
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DOMAIN	STATUS	GEOLOCATION
android.googleusercontent.com	ok	IP: 74.125.131.82 Country: United States of America Region: California City: Mountain View Latitude: 37.405991 Longitude: -122.078514 View: Google Map

SCAN LOGS

Timestamp	Event	Error
2026-04-25 13:46:14	Generating Hashes	OK
2026-04-25 13:46:14	Extracting APK	OK
2026-04-25 13:46:14	Unzipping	OK
2026-04-25 13:46:14	Parsing APK with androguard	OK
2026-04-25 13:46:15	Extracting APK features using aapt/aapt2	OK

2026-04-25 13:46:15	Getting Hardcoded Certificates/Keystores	OK
2026-04-25 13:46:17	Parsing AndroidManifest.xml	OK
2026-04-25 13:46:17	Extracting Manifest Data	OK
2026-04-25 13:46:17	Manifest Analysis Started	OK
2026-04-25 13:46:17	Performing Static Analysis on: Uncrackable Level 2 (owasp.mstg.uncrackable2)	OK
2026-04-25 13:46:18	Fetching Details from Play Store: owasp.mstg.uncrackable2	OK
2026-04-25 13:46:18	Checking for Malware Permissions	OK
2026-04-25 13:46:18	Fetching icon path	OK
2026-04-25 13:46:18	Library Binary Analysis Started	OK
2026-04-25 13:46:18	Analyzing lib/armeabi-v7a/libfoo.so	OK
2026-04-25 13:46:18	Analyzing lib/x86/libfoo.so	OK

2026-04-25 13:46:18	Analyzing lib/arm64-v8a/libfoo.so	OK
2026-04-25 13:46:18	Analyzing lib/x86_64/libfoo.so	OK
2026-04-25 13:46:18	Analyzing apktool_out/lib/armeabi-v7a/libfoo.so	OK
2026-04-25 13:46:18	Analyzing apktool_out/lib/x86/libfoo.so	OK
2026-04-25 13:46:18	Analyzing apktool_out/lib/arm64-v8a/libfoo.so	OK
2026-04-25 13:46:18	Analyzing apktool_out/lib/x86_64/libfoo.so	OK
2026-04-25 13:46:18	Reading Code Signing Certificate	OK
2026-04-25 13:46:19	Running APKiD 3.0.0	OK
2026-04-25 13:46:20	Detecting Trackers	OK
2026-04-25 13:46:20	Decompiling APK to Java with JADX	OK
2026-04-25 13:46:27	Converting DEX to Smali	OK

2026-04-25 13:46:27	Code Analysis Started on - java_source	OK
2026-04-25 13:46:27	Android SBOM Analysis Completed	OK
2026-04-25 13:46:59	Android SAST Completed	OK
2026-04-25 13:46:59	Android API Analysis Started	OK
2026-04-25 13:47:31	Android API Analysis Completed	OK
2026-04-25 13:47:31	Android Behaviour Analysis Started	OK
2026-04-25 13:47:33	Android Behaviour Analysis Completed	OK
2026-04-25 13:47:33	Extracting Emails and URLs from Source Code	OK
2026-04-25 13:47:33	Email and URL Extraction Completed	OK
2026-04-25 13:47:33	Extracting String data from APK	OK
2026-04-25 13:47:33	Extracting String data from SO	OK

2026-04-25 13:47:33	Extracting String data from Code	OK
2026-04-25 13:47:33	Extracting String values and entropies from Code	OK
2026-04-25 13:47:33	Performing Malware check on extracted domains	OK
2026-04-25 13:47:34	Saving to Database	OK

Report Generated by - MobSF v4.5.0

Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

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